

WASP-11b

R-0883

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-10_181016 · created 2026-07-29T00:57:46+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458407.7169 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.157 +/- 0.051
Transit depth (Rp/Rs)^2	2.47 +/- 1.61 [%]
Orbital Inclination (inc)	84.56 +/- 1.72
Airmass coefficient 1 (a1)	0.77 +/- 0.5
Airmass coefficient 2 (a2)	0.069 +/- 0.063
Scatter in the residuals of the lightcurve fit is	60.2 %
Best Comparison Star	None
Optimal Aperture	1.82
Optimal Annulus	3.0
Transit Duration (day)	0.034 +/- 0.038

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.157 ± 0.051 vs 0.131 (deviation 0.39σ)
Duration fit vs published	0.034 d vs 0.1075 d (deviation 1.49σ)
Mid-time O–C	-0.1 min
Depth SNR	2.4
Residual scatter	60.2 %

Light curve

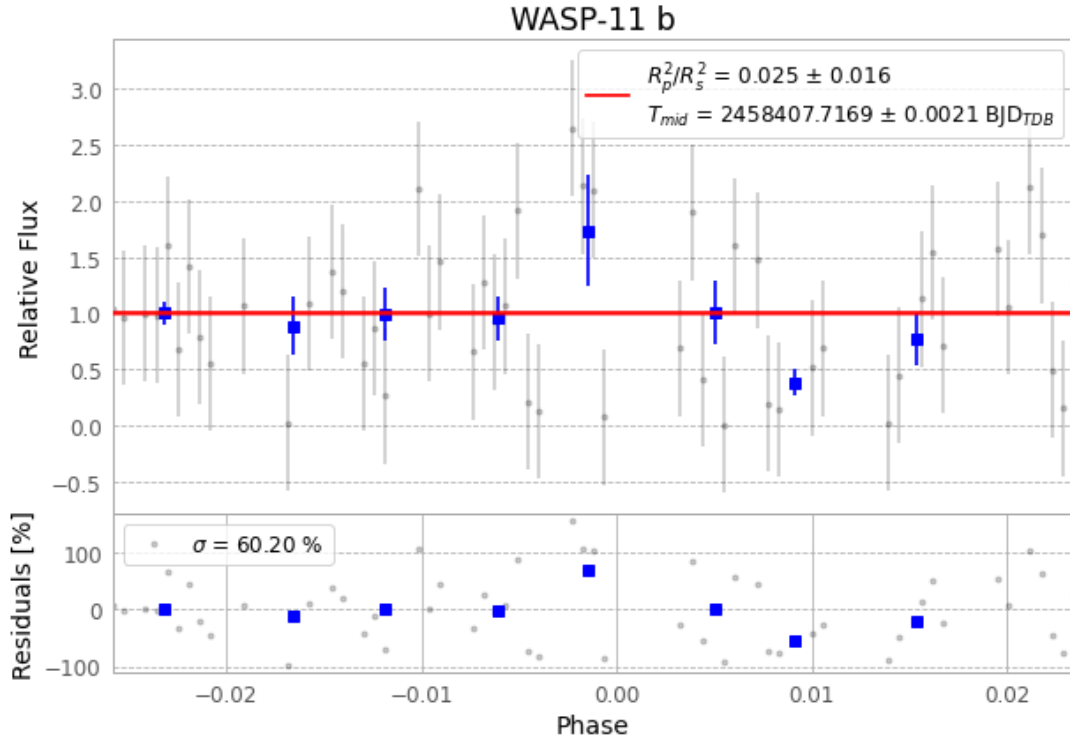


Figure 1 — FinalLightCurve_WASP-11 b_2018-10-15.png (SHA-256 4181e7e5163831b5...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [338, 447], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 527eae67e031e6ce...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

92 FITS frames (60 MB), HATP-10181016024212.FITS → HATP-10181016071512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-181016.log (1f384f6ab0e4...), FinalLightCurve_WASP-11 b_2018-10-15.png (4181e7e51638...), FinalLightCurve_WASP-11 b_2018-10-15.csv (fbcadafc1f1d...)

Compute resources

Wall time 140.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-29 00:57Z.